

第二章 解析函数

Here are some complex functions:

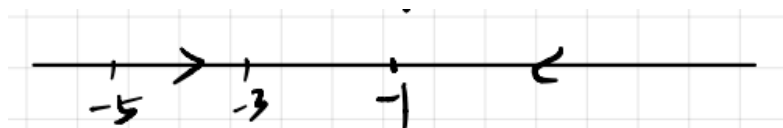
$$z + 1, \quad z^2, \quad 2z^2 + z - 1, \quad e^z, \quad e^{5z^2+2}$$

$$\cos z, \quad \cos(2z^2 + z - 1), \quad \ln z, \quad z_1^{z_2}, \quad \frac{P(z)}{Q(z)}, \quad \cosh z, \quad \sinh z$$

2.1 Limit

$$f(x) \quad x \rightarrow x_0 \quad |x - x_0| \rightarrow 0$$

On the real domain, the limit is approximated from two or one direction.



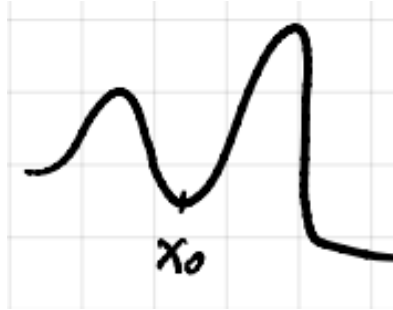
On the complex domain (2-D), the limit is approximated from multiple directions.



2.2 Continuous function

$$x \rightarrow x_0$$

$$f(x) \rightarrow f(x_0)$$



Then we can generalize it to the complex domain. If a function is continuous at point z_0 , then $f(z) \rightarrow f(z_0)$ when $z \rightarrow z_0$.

$$\lim_{z \rightarrow z_0} f(z) = f(z_0)$$

Definition: the limit of continuous complex functions:

If $|f(z) - f(z_0)| \rightarrow 0$, when $|z - z_0| \rightarrow 0$.

$$\lim_{z \rightarrow z_0} f(z) = f(z_0)$$

3. Derivative

Real function

A real function $f(x)$ has a derivative at x :

$$f'(x) \triangleq \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

Example:

$$\lim_{\Delta x \rightarrow 0} \frac{(x_1 + \Delta x)^2 - x^2}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{x^2 + 2x \cdot \Delta x + \Delta x^2 - x^2}{\Delta x} = \lim_{\Delta x \rightarrow 0} (2x + \Delta x) = 2x$$

Now, let's move to complex functions. The derivative of $f(z)$ at z_0 is:

$$f'(z) \triangleq \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}$$

Example 1:

$$f(z) = z^2$$

$$f'(z) = \frac{z^2 + 2z \cdot \Delta z + (\Delta z)^2 - z^2}{\Delta z} = 2z \cdot \Delta z + (\Delta z)^2$$

Example 2:

$$f(z) = z^2, \quad z = x + iy$$

$$f(z) = (x + iy)^2 = x^2 - y^2 + i(2xy) = u(x, y) + iv(x, y)$$

Example 3:

$$\begin{aligned} f(z) &= \frac{1+z}{1-z} = \frac{1+x+iy}{1-x-iy} \\ &= \frac{(1+x+iy)(1-x+iy)}{(1-x)^2 + y^2} = \frac{(1+iy)^2 - x^2}{(1-x)^2 + y^2} \\ &= \frac{1-y^2 + i2y - x^2}{(1-x)^2 + y^2} \end{aligned}$$

Thus, we have:

$$u(x, y) = \frac{1-y^2-x^2}{(1-x)^2 + y^2}, \quad v(x, y) = \frac{2y}{(1-x)^2 + y^2}$$

With the above knowledge, we can easily solve the following equations:

$$\begin{aligned} \int_{-\infty}^{+\infty} \frac{\sin x}{x} dx &= \pi, \quad \int_{-\infty}^{+\infty} \frac{\sin^4 x}{x^4} = \frac{2\pi}{3} \\ \int_0^{\infty} \frac{\ln x}{1+x+x^2} dx, \quad \sum_{i=1}^{\infty} \frac{1}{n^2} &= \frac{\pi^2}{6} \end{aligned}$$

4. Cauchy-Riemann condition

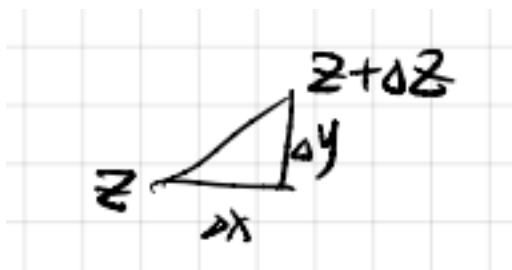
A function is analytic if it satisfies the Cauchy-Riemann conditions:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\Delta z = \Delta x + i\Delta y$$

$$\begin{aligned} f'(z) &= \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} \\ &= \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{k(x + \Delta x, y + \Delta y) + iv(x + \Delta x, y + \Delta y) - u(x, y) - iv(x, y)}{\Delta x + i\Delta y} \end{aligned}$$

Δz 可以从多个方向发生逼近，我们关注两个特殊的方向。



(1) 水平逼近

$$\begin{aligned}
 & \Delta y = 0 \\
 &= \lim_{\Delta x \rightarrow 0} \frac{u(x + \Delta x, y) + iv(x + \Delta x, y) - u(x, y) - iv(x, y)}{\Delta x} \\
 &= \lim_{\Delta x \rightarrow 0} \frac{u(x + \Delta x, y) - u(x, y)}{\Delta x} + i \lim_{\Delta x \rightarrow 0} \frac{v(x + \Delta x, y) - v(x, y)}{\Delta x} \\
 &= \frac{\partial u(x, y)}{\partial x} + i \frac{\partial v(x, y)}{\partial x}
 \end{aligned}$$

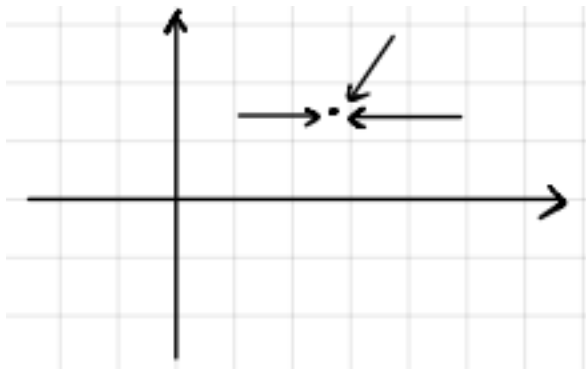
(2) vertical path

$$\begin{aligned}
 & \Delta x = 0 \quad \Delta z = i\Delta y \\
 & \lim_{\Delta y \rightarrow 0} \frac{u(x, y + \Delta y) + iv(x, y + \Delta y) - u(x, y) - iv(x, y)}{i\Delta y} \\
 &= \lim_{\Delta y \rightarrow 0} \frac{u(x, y + \Delta y) - u(x, y)}{\Delta y} \cdot (-i) + \lim_{\Delta y \rightarrow 0} \frac{v(x, y + \Delta y) - v(x, y)}{i\Delta y} \cdot i \\
 &= \frac{\partial u}{\partial y}(-i) + \frac{\partial v}{\partial y}
 \end{aligned}$$

然后我们可以得到 **the Cauchy-Riemann condition.**

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

我们还可以从其他方向逼近，如下图所示。



在这种情况下， $\Delta z = \Delta x + i\lambda\Delta x$ 。如果实部和虚部不独立，the Cauchy-Riemann condition 成立, 并且 $f(z)$ 可解析。

即使实部和虚部都有导数，函数也可能不是解析的。

Example 1

(1) $f(z) = x + iy$, 如果 $f(z)$ 可解析，那么我们一定可以得到 the Cauchy-Riemann condition 成立

$$\begin{aligned}\frac{\partial u}{\partial x} &= \frac{\partial v}{\partial y} = 1 \\ \frac{\partial u}{\partial y} &= \frac{\partial v}{\partial x} = 0\end{aligned}$$

(2) $f(z) = 2x + iy$

$$\frac{\partial u}{\partial x} = 2, \frac{\partial v}{\partial y} = 1$$

该函数不可解析。由于相同原因, $f(z) = x^2 + iy$ 也不可解析。

The Cauchy-Riemann condition 是函数可解析的必要条件, 所以我们可以用 the Cauchy-Riemann condition 通过反证法证明函数不可解析。

(3) $f(z) = z^2 = x^2 - y^2 + i2xy$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = 2x, \quad \frac{\partial u}{\partial y} = -2y = -\frac{\partial v}{\partial x}$$

该函数可解析。

Example 2

$f(z)$ is analytic, its real part is $v(x, y) = x^2 - y^2$, find its $v(x, y)$.

$$\begin{aligned}\frac{\partial u}{\partial x} &= \frac{\partial v}{\partial y} \Rightarrow 2x = \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} &= -\frac{\partial v}{\partial x} \Rightarrow -2y = -\frac{\partial v}{\partial x} \\ &\Rightarrow v(x, y) = 2xy + c\end{aligned}$$

5.二阶导数

$u(x, y)$ 的二阶导数

$$\frac{\partial^2 u}{\partial x^2} \quad \frac{\partial^2 u}{\partial y^2} \quad \frac{\partial^2 u}{\partial x \partial y}$$

利用 the Cauchy-Riemann condition, 我们可以得到

$$\begin{aligned}\frac{\partial^2 u}{\partial x^2} &= \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right) = \frac{\partial}{\partial x} \left(\frac{\partial v}{\partial y} \right) = \frac{\partial^2 v}{\partial x \partial y} \\ \frac{\partial^2 u}{\partial y^2} &= \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial y} \right) = \frac{\partial}{\partial y} \left(-\frac{\partial v}{\partial x} \right) = -\frac{\partial^2 v}{\partial x \partial y}\end{aligned}$$

利用上面两个方程, 我们可以得到 **Laplace equation / Harmonic condition.**

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0u_{xx} + u_{yy} = 0$$

函数可解析 $\Rightarrow$ the Cauchy-Riemann condition $\Rightarrow$ the Laplace equation

6.复变函数的导数公式

$$\begin{aligned}(z^2)' &= 2z \\ (z^n)' &= n \cdot z^{n-1}\end{aligned}$$

$\frac{P(z)}{Q(z)}, (e^z)' = e^z$ 导数和实变函数一样。

proof

$$\begin{aligned}(e^z)' &= \frac{e^{z+\Delta z} - e^z}{\Delta z} = \frac{e^{x+\Delta x+i(y+\Delta y)} - e^{x+iy}}{\Delta x + i\Delta y} \\ &= \frac{e^{x+\Delta x}[\cos(y+\Delta y) + i\sin(y+\Delta y)] - e^x[\cos y + i\sin y]}{\Delta x + i\Delta y} \\ &= e^z\end{aligned}$$

考虑三角函数

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}$$

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}$$

在实数域, $(\sin x)' = \cos x$, 在复数域, $(\sin z)' = \cos z$ 也是这样的形式, 因为实数也是复数。

proof

$$(\sin z)' = \frac{1}{2i} [e^{iz} \cdot i - e^{iz}(-i)] = \cos z$$

$$(\cos z)' = \frac{1}{2} (e^{iz}i - ie^{-iz}) = -\sin z$$

考虑双曲函数

$$\cosh z = \frac{e^z + e^{-z}}{2} \quad \sinh z = \frac{e^z - e^{-z}}{2}$$

$$(\sinh z)' = \cosh z \quad (\cosh z)' = \sinh z$$